

Modular Arithmetic Problems

1. What is the remainder when $6^{273} + 8^{273}$ is divided by 49?
2. The cube of any integer is of the form:
a. $9K$ b. $9K+1$ c. $9K+8$ d. All of these
3. Find the remainder of the number $2005^{2007^{2009}}$ when divided by 7.
4. What is the last digit of $\left((7^7)^7 \dots\right)^7$ if there are one thousand 7s as exponents? There is only one 7 in the middle.
5. Can you find a number that is both a multiple of 2 but not a multiple of 4 and a perfect square?
6. If 'n' is a number that is made from a string of 5s and is divisible by 2003. What are the last 6 digits of the quotient when 'n' is divided by 2003?
7. Find the remainder when the number $10^{10} + 10^{100} + 10^{1000} + \dots + 10^{10000000000}$ is divided by 7.
8. Find the last two digits of 2^{999} .
9. Find the last two digits of $(2026)^{2026}$.
10. Prove that a power of 2 cannot end with four equal digits.
11. Let $n = 16^{3^r} - 4^{3^r} + 1$; for some positive integer r . Prove that $2^{n-1} - 1$ is divisible by n .
12. Find the smallest odd prime factor of $20198 + 1$.
13. For all $a, n \in \mathbb{N}$, solve $a^{n+1} - (a+1)^n = 2001$.
14. Find all $n \in \mathbb{N}$, such that $7 \mid 2^n - 1$. Also prove that there is no $n \in \mathbb{N}$, with $7 \mid 2^n + 1$.
15. Find the smallest positive integer K , such that $2^{69} + K$ is divisible by 127.

Resources:

1. [Foundation of Divisibility and Congruence: Lecture Notes](#)
2. [104 Number Theory Problems: Titu Andreescu, Dorin Andrica, Zuming Feng.](#)
3. [Olympiad Number Theory: Justin Stevens.](#)
4. [Number Theory: Michael Penn.](#)
5. [Number Theory Olympiad Problems Discussion.](#)